# Active Maintenance Report: coppock_kaur_2022

**Coppock and Kaur (2022, American Journal of Political Science): Qualitative Imputation of Missing Potential Outcomes**

2026-08-03

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**Archived at Zenodo:** doi:10.5281/zenodo.24168367. This DOI always resolves to the latest release of the maintained version.

*Drafted by Claude Opus 5 under the supervision of Alex Coppock.*

This repository holds the actively maintained replication code for Coppock and Kaur (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

| | |
|---|---|
| Article | 10.1111/ajps.12697 |
| Replication archive | 10.7910/DVN/2IVKXD |

**The data are not redistributed here.** The deposit is 10 files and about 0.7 MB, and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

**Repository layout.** `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the exhaustive list of numbers the article and its supporting information print, the ground truth CSV is the comparison against the deposit and against the rewrite, and `build_ground_truth.R` is the gate that stops the run if a published number is checked by neither instrument; it also writes `float_coverage.csv`, a count of the numbers each published table and figure prints and how many of them are reproduced. `run_archive.R` runs the deposited scripts on their own and records the outcome of each in `archive_run_status.csv` and every file a run writes in `archive_overwrites.csv`; it is not part of `run_all.R`, because it needs the deposit's own packages. `coppock_kaur_2022_errata.pdf` at the root lists the numbers in the article that its own tables, appendix and data do not support. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

**License.** CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

**To reproduce.** Clone or download the repository, open `coppock_kaur_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 10 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, janitor, here, and for this report knitr and kableExtra. Paths resolve through `here`, so nothing depends on the working directory. Almost all of the run time is in the last two scripts, which simulate the bounds twenty-one times over in order to say what the deposited archive's method gives; the analysis itself is quick. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The figure PDFs always differ, because a PDF records the time it was written; the CSV and PNG output comes back byte-identical.

# 1 Summary

Two questions, answered before the detail.

## 1.1 Does the deposited archive run?

Yes. All five analysis scripts and the helpers file execute without error on R 4.6.0. Every package the archive names still installs from CRAN: tidyverse, xtable, janitor and reshape2. Nothing in the archive has been removed from the language; what has accumulated is deprecation. `geom_errorbarh()` and the `<ggplot> %+% data` idiom were deprecated in ggplot2 4.0.0, the `size` aesthetic for lines in 3.4.0, and `reshape2::melt` and `dcast` are superseded by tidyr. All of them still work and all of them warn. Everything that stands between this archive and a clean run is resolvable by substitution.

Running is not the same as reproducing, and one thing keeps the archive from reproducing its own figures exactly. `figures_2_3_A1_A2.R` estimates the bounds by simulation, 1,000 draws for Figures 2 and A1 and 10,000 for Figures 3 and A2, and it sets no seed. The published figures are labelled with rounded percentage points, so most of the simulation noise is absorbed by the rounding, but not all of it: repeating that simulation at twenty seeds moves at least one of the 70 labels the four figures determine off the value the bound actually takes in 14 of them. A reader running the deposited script today has better than even odds of seeing a figure that disagrees with the article somewhere, with no error and no warning. The maintained rewrite does not simulate these bounds at all, for the reason set out below.

Two smaller things. `table_4.R` prints its bounds as fractions where the paper prints percentage points, so the deposited output reads `[0, 0.3]` against the article's `[0, 30]`. It is a display inconsistency and not a numerical one. And the writing of every table and figure is commented out: each script builds the object and prints it, but the `xtable` and `ggsave` calls that would put it on disk are behind `#`. The archive is a set of scripts that show their work rather than a pipeline that produces the article's floats.

### 1.2 Does the maintained rewrite reproduce the paper?

Mostly, and the exceptions have one cause. 201 of the 227 verifiable ground truth claims match the published values to reported precision: every cell and bound of Table 3, every bound of Figure 1, every row of Table 4, the democratization row of Table 5, every label of Figures A1 and A2, all ten bounds of Figure 4, 761 of the 763 cells the two appendix case tables print, the abstract, and every in-text quantity from the Expert Survey section and Appendix A. The article's headline result is among them: the extreme value bounds on the ATE come to [-2, 49], 51 points wide, as published. A few of those matching labels are printed here at one decimal rather than as a whole number, because the bound falls exactly halfway between two whole numbers and no whole number states it; each agrees with the published label at the precision the page prints, and the table under *The unseeded simulation* lists every one.

The 26 that do not match fall into three groups.

17 of them follow from a single defect in the deposited data, described in full below: 7 imputation probabilities in `cases_clean.csv` do not match the probability the article's own appendix C states for that case, and the rewrite uses the stated values. Four are labels of Figure 2, twelve are labels of Figure 3, and one is the sentence summarising the ATT.

8 are places where the article contradicts itself. Seven of them are numbers in prose, spread over five sentences, two of which state a pair of numbers apiece; the rewrite agrees with the table or the appendix the sentence disagrees with in every case. The eighth is a cell of appendix Table B.2, an imputation probability the table prints differently from both appendix C and the deposited data.

The last is a second cell of appendix Table B.2, an observed outcome the deposited data record differently.

The remaining 5 recorded quantities have no comparison to make: four are the end-of-conflict row of the imputation summary, which the rewrite computes and neither the article nor the appendix prints, and one is a survey fact the deposit never recorded.

All of it is set out with corrected sentences and corrected floats in `coppock_kaur_2022_errata.pdf` at the root of this repository. **No conclusion of the article changes.** One published quantity moves materially: the average treatment effect on the treated, which the article states as a -15 percentage point effect and which is -22.5 points once the imputation probabilities appendix C states are the ones the analysis uses.

## 2 Paper overview

**Citation**: Coppock, A. and Kaur, D. (2022). "Qualitative imputation of missing potential outcomes." *American Journal of Political Science*, 66(3), 681-695. DOI: 10.1111/ajps.12697

**Summary**: The paper proposes a framework for meta-analysing qualitative causal inferences. A causal claim about a single case is a claim about the value of a missing potential outcome, so a set of such claims can be treated as imputations and fed into Manski (1999) extreme value bounds, which are the logical range of average causal effects consistent with what has been observed. Bounds on a binary outcome start 200 percentage points wide and narrow to 100 once the world reveals half the potential outcomes; each qualitative imputation narrows them further, and cases too murky to call are left unimputed rather than guessed. Where the analyst holds a probability rather than a point belief, the extension enumerates the 2^k consistent worlds and takes the probability-weighted bounds. The application is to 63 cases eligible for a transitional truth commission upon democratization, eight of which received one, with authoritarian resumption within ten years as the outcome. Imputation in four steps takes the bounds on the ATE to [-2, 49], 51 points wide. An email survey of country experts, 20 of whom responded, supplies an independent set of imputations; deferring to the experts where they answered narrows the bounds to 44 points. Appendix A repeats the exercise for 54 end-of-conflict cases.

# 3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

| Script | Status on current R | Resolution |
|---|---|---|
| quimpo_helpers.R | Clean (sourced by the analysis scripts) | %>% to |>, map_df() to map() + bind_rows() |
| table_3_figure_1.R | Runs; deprecation warning | size= to linewidth= |
| table_4.R | Runs clean | Multiply the bounds by 100 to print percentage points, as the paper does |
| table_5.R | Runs clean | recode_factor() to case_when() + factor() |
| figures_2_3_A1_A2.R | Runs; three deprecation warnings; unseeded simulation | melt/dcast to pivot_longer/pivot_wider; geom_errorbarh() to geom_errorbar(orientation='y'); %+% to a named plot function; do() to reframe(); the bounds enumerated rather than simulated |
| figure_4.R | Runs; deprecation warning and a melt() id-variable warning | melt to pivot_longer with an explicit names_to; size= to linewidth= |

## 3.1 The unseeded simulation, and the bounds it was estimating

The probabilistic extension is what makes Figures 2, 3, A1 and A2 stochastic in the archive. Where the authors hold a probability rather than a point belief about a missing potential outcome, the deposit's `sample_bounds()` draws binary potential outcomes from those probabilities and recomputes the bounds, and the reported estimate is the mean over draws. What the archive omits is `set.seed()`, so its figures are one unlabelled draw.

**The maintained rewrite computes these bounds exactly, where the deposit estimates them by simulation.** A bound is affine in the realised imputations: it is the count of realised ones among the imputed and observed entries, offset by the unimputed entries the bound fills with a constant, over the number of units. So the distribution the deposit samples from can be written down and enumerated instead, by convolving one Bernoulli imputation in at a time, and `bounds_distribution()` in `maintained/helpers.R` does that. The plotted estimate is the mean of that distribution and the interval its 2.5th and 97.5th quantiles.

This is not a different estimator. It is the same quantity by a different route, and it is the route the paper itself takes for the toy example, whose Table 4 enumerates all four consistent worlds and takes the probability-weighted bounds rather than sampling them. What it changes is threefold. Nothing in this repository depends on a seed, so the reproduction check is a byte comparison rather than a judgment about noise. The 84 bounds the four figures carry are now stated at their exact values rather than to within the simulation's error, which at the deposit's number of draws reaches 0.33 points at one of them. And a bound that falls exactly halfway between two whole numbers becomes visible as such, instead of being settled by whichever side a draw happened to land on.

`maintained/text_expected_bounds.R` keeps the comparison against the deposit's own method on the record: it runs that simulation once at a fixed seed, writes what it gives beside the exact value, and stops the build if any of the 84 bounds disagree by more than five standard errors of the simulated estimate. `maintained/text_seed_sensitivity.R` runs it at twenty seeds.

Table 2: The deposited archive's method, repeated at twenty seeds. Of the labels the four figures carry, the number the simulation rounds to a different whole number than the bound it is estimating does.

| Labels rounded differently | Seeds |
|---:|---:|
| 0 | 6 |
| 1 | 8 |
| 2 | 3 |
| 3 | 3 |

The size of the disturbance is one percentage point in the label, never more, and it barely touches the substantive claims: the final democratization bounds read [-2, 49] at 17 of the 20 seeds and [-1, 49] at the other 3, a difference of one point in the lower label. The point is not that the archive gets a different answer but that it gets a slightly different figure each time it is run, and it offers a reader no way to tell whether a discrepancy is noise or a mistake.

The table above counts only the 70 labels a whole number can state. The other 14 fall exactly halfway between two of them, and for those the deposit's method prints whichever side the draw fell on. The maintained figures print such a bound at the one decimal that states it.

Table 3: Bounds falling on a whole-number rounding boundary, with what one run of the deposit's method gives for the same quantity. All but the democratization ATT rows from Treated Cases onwards are a property of the deposit as published; those appear once the appendix C probabilities are restored.

| Sample | Estimand | Step | Exact bound | Simulated, at seed 12345 |
|---|---|---|---|---|
| Democratization | ATT | Initial Values | [-87.5, 12.5] | [-88, 12] |
| Democratization | ATT | Disbanded and Discredited Cases | [-87.5, 12.5] | [-88, 12] |
| Democratization | ATT | Treated Cases | [-22.5, -22.5] | [-22, -22] |
| Democratization | ATT | Non-transitional Cases | [-22.5, -22.5] | [-23, -23] |
| Democratization | ATT | Untreated Cases | [-22.5, -22.5] | [-22, -22] |
| Democratization | ATT | Unimputable Cases | [-22.5, -22.5] | [-22, -22] |
| End of conflict | ATU | Disbanded and Discredited Cases | [-27.5, 64.2] | [-27, 64] |
| End of conflict | ATU | Treated Cases | [-27.5, 64.2] | [-28, 64] |

The 4 democratization ATT rows from *Treated Cases* onwards are the consequential ones, and they are the only rows in the table the corrections create. The ATT is exactly -22.5 once South Africa carries the probability appendix C states for it, so a whole-number label for it is not decided by the estimate at all: over twenty seeds the deposit's method prints [-22, -22] at 12 and [-23, -23] at 8. The rest of the table was on a boundary in the deposit as published, and the labels the article prints for them are one draw of a coin flip.

## 3.2 Checksums

Table 4: The deposit as Dataverse serves it with ?format=original. All ten files match the checksums Dataverse publishes, and all ten match the copy this program downloaded in March 2026.

| File | Bytes | MD5 served (first 12) | Published MD5 agrees |
|---|---|---|---|
| cases_clean.csv | 21433 | 0a238c073307 | yes |
| codebook_cases_clean.pdf | 348561 | 87154446febf | yes |
| figure_4.R | 4282 | d62dbb3dc25a | yes |
| figures_2_3_A1_A2.R | 5859 | b06b1c39bfba | yes |
| logfile.html | 278521 | 831c13a4b09c | yes |
| quimpo_helpers.R | 1517 | 876ec89e03d6 | yes |
| README.txt | 1712 | 634560b77b47 | yes |
| table_3_figure_1.R | 3456 | e8fb68e7b478 | yes |
| table_4.R | 2220 | 237c27908efb | yes |
| table_5.R | 1184 | fd0efb3ad0e6 | yes |

Harvard Dataverse ingests `cases_clean.csv` as tabular data and derives a `.tab` representation from it, so the manifest records `originalFileName` and `originalFileSize` rather than the derived names and sizes, and `download_original.R` asks for `?format=original`. The published MD5 describes the deposited bytes, not the derived ones, and here it verifies them

exactly. Other deposits in this program carry published checksums that verify neither file, so the download script checks against the served checksum and reports any published disagreement rather than failing on it.

# 4 Errata

Nothing in the deposited code is wrong. The deposited data are wrong in seven cells, and eight claims in the article's prose, spread over six sentences, are wrong on the article's own terms. All of it is set out with corrected sentences and corrected floats as twenty-two numbered entries in `coppock_kaur_2022_errata.pdf` at the root of this repository. None of it changes a conclusion. The sections below carry the note's numbering.

## 4.1 Entries 1 and 2: seven imputation probabilities the deposit transcribes differently from appendix C

The imputed potential outcomes in this article are not measurements. They are qualitative judgments about what would have happened in each case, set out one by one in appendix C with the scholarship behind each of them, and the deposited `cases_clean.csv` is a transcription of those judgments. Appendix C states an imputation probability for 30 of the 31 imputed democratization cases; the exception, Thailand (1991-1992), refers to the entry above it rather than stating a number of its own. Twenty-three of the 30 agree with the deposit exactly. 7 do not.

Table 5: The imputation probabilities the deposited case file transcribes differently from the appendix C narrative that states them. The imputed value is the point imputation, which agrees with appendix C in every case.

| Case | Appendix C entry | Imputed value | Probability in the deposit | Probability in appendix C |
|---|---|---|---|---|
| South Africa (1910-1994) | C.2.7 | 1 | 0.2 | 0.8 |
| Ghana (1981-1993) | C.3.2 | 0 | 0.1 | 0.2 |
| Uruguay (1973-1984) | C.3.6 | 1 | 1 | 0.9 |
| Sierra Leone (1997-1998) | C.4.8 | 0 | 0.2 | 0.1 |
| Nicaragua (1979-1990) | C.4.10 | 1 | 0.8 | 0.9 |
| Central African Rep (1981-1993) | C.4.12 | 1 | 0.3 | 0.7 |
| Sierra Leone (1992-1996) | C.4.13 | 1 | 0.3 | 0.7 |

3 of the 7 are exact complements, and in all three the deposit contradicts itself: an imputed potential outcome of 1 sits beside a probability below one half, which says the imputation is more likely to be 0 than 1. The main text states the South Africa probability too, on page 10, and states it as 0.8. The other 4 differ from the stated probability by one step of 0.1 and none of them crosses one half, so none changes an imputed value.

Appendix Table B.2 prints the deposited value in all 7 cases, so the published table inherits the transcription error rather than committing a second one. `maintained/table_b1_b2_case_dataset.R` is the one script in the rewrite that reads the deposit rather than the corrected file, precisely so that this stays checkable: the question those two tables answer is whether the published pages print what the deposit holds, and they do.

**The rewrite uses the values appendix C states.** `maintained/apply_appendix_c_corrections.R` is a visible, named step that writes both the corrected case file every analysis script reads (as `output/cases_corrected.rds`, and as `output/cases_corrected.csv` for anyone who wants the corrected data without running R) and `output/appendix_c_corrections.csv`, the table above, so the deposited value sits beside the value that replaced it. It asserts the deposited value in every column it touches before overwriting it, so a change to the deposit stops the run rather than being absorbed. The probability columns are cumulative, and each correction is applied to every step column from the step at which the case is imputed onwards.

### 4.1.1 What it moves

The article's headline result is unchanged. The extreme value bounds on the ATE stay at [-2, 49] and 51 points wide, as do the abstract's three numbers, the expert survey comparison in Figure 4, Table 5, and every quantity in the end-of-conflict application, all seven corrections being democratization cases. The two intermediate steps of Figure 2 move by about one point in each label, and the ATU bounds of Figure 3 by about one and a half.

The consequential movement is in the ATT. Every treated case has both potential outcomes filled in by the second imputation step, which is why the ATT bounds collapse to a point, and South Africa is one of the eight treated cases. Restoring its probability from 0.2 to the 0.8 that both the main text and appendix C state moves the ATT from -15 to -22.5 percentage points.

Table 6: Every published bound the corrections move. Published values are the whole-number labels the figures carry; corrected values are exact. The third row is the article's headline result and rounds to the published label.

| Quantity | Published | Corrected |
|---|---|---|
| Figure 2, Treated Cases | [-16, 68] | [-16.8, 67.3] |
| Figure 2, Non-transitional Cases | [-12, 62] | [-13.2, 61.4] |
| Figure 2, Untreated and Unimputable Cases | [-2, 49] | [-1.6, 49.2] |
| Figure 3, ATU, Untreated and Unimputable Cases | [0, 58] | [1.5, 59.6] |
| Figure 3, ATT, Treated Cases onwards | [-15, -15] | [-22.5, -22.5] |

The article's ATU claim is untouched. It says the ATU bounds are "greater than 58 points wide", and that width is 58.2 points before the correction and after it, because both ATU bounds move by the same amount.

One imputation changes class. An imputation is made with certainty when its probability is exactly 0 or exactly 1, and Uruguay (1973-1984) is carried at 1 in the deposit where appendix C entry C.3.6 states 0.9. The split the article states as "5 with certainty, 26 probabilistically" is 4 and 27 in the deposit, Uruguay being the difference, and 3 and 28 once appendix C is followed. The number left unimputed, 32, and the 31 imputed in total are the same either way.

#### 4.1.2 One inconsistency this does not resolve

South Africa - Namibia (1966-1988), an end-of-conflict case, is the fourth place in the deposited data where a point imputation sits on the wrong side of its own probability: the imputed treated outcome is 0 and the probability that it equals 1 is 0.8. Appendix Table B.1 prints the same pair, so the table and the deposit agree. Appendix C covers the democratization cases only and Appendix A gives no case-by-case narrative for the end-of-conflict sample, so there is no stated probability anywhere to resolve it against, and the case is left exactly as deposited. The three democratization cases of the same shape are corrected above, because for those a stated probability exists.

### 4.2 Entries 3 and 4 through 10: eight claims in the article's prose

**Entry 5, the toy example's treated outcomes.** The text introducing the toy example says "the outcome for three of the treated units and one of the untreated units is 1". Table 3 on the facing page shows units 1 through 4 treated with an observed outcome of 1, which is four, and the deposited `Y <- rep(c(1, 0, 1, 0), c(4, 3, 1, 2))` encodes four. Four is also what the bounds require: the initial bounds of [-40, 60] follow from four treated ones and would be [-50, 50] with three.

**Entry 6, the probabilistic extension's probabilities.** The text says that instead of imputing 0 for units 6 and 7, "we think the probabilities of being a '1' for units 6 and 7 are .3 and .4". Table 4 gives the four scenario probabilities as 0.56, 0.14, 0.24 and 0.06, which is the product distribution of 0.2 and 0.3, not of 0.3 and 0.4, which would give 0.42, 0.18, 0.28 and 0.12. The deposited `table_4.R` uses 0.2 and 0.3 and reproduces Table 4 exactly, as does the rewrite.

**Entry 7, Appendix A's bounds width.** Appendix A says the imputations shrink "the width of the extreme value bounds from 100 to 41 points" and then, in the next sentence, that "the bounds come to [-3, 42] (only 45 points wide)". Both cannot hold. The pipeline gives final end-of-conflict bounds of [-2.9, 41.5], a width of 44.4 points, which rounds to 44 and which is 45 if taken as the difference of the rounded endpoints, as the second sentence does. The 41 matches nothing.

**Entry 4, how the 63 missing potential outcomes were imputed.** The empirical section says "for all 63 unobserved potential outcomes, we imputed 5 with certainty, 26 probabilistically

and we left 32 unimputed". Neither number holds under either reading of the data, as set out above. The 32 left unimputed and the 31 imputed in total are right.

**Entry 8, the count of disbanded and discredited cases.** Step 1 names two democratization cases, Bolivia and the Philippines, then says in the next sentence that "the observed outcome $Y_i(0)$ in each of these four cases was 0". Appendix Table B.2 lists two cases at that step. Appendix Table B.1 lists four end-of-conflict cases at the same step, which is where the count appears to have come from.

**Entry 3, the ATT.** "We can summarize the ATT as a -15 percentage point effect on return to authoritarianism." That is what the deposited data give; it is -22.5 under the appendix C probabilities.

### 4.3 Two appendix cells the deposit does not support, and two it transposes

Appendix Tables B.1 and B.2 print the full case datasets, 763 cells between them, and the deposited case file reproduces 761.

**Burundi's imputation probability.** Table B.2 prints the imputed treated outcome for Burundi (1996-2005) as 1 with a probability of 0.9. Appendix C entry C.4.11 states 0.8 for the same case, and the deposited data carry 0.8. Deposit and appendix agree here, so there is nothing to correct in the data; the table cell is the outlier and no bound is affected.

**Azerbaijan's observed outcome (entry 11).** Table B.2 prints an observed outcome of 0 for Azerbaijan (1991-1992), consistent with the untreated potential outcome of 0 printed beside it. The deposited `outcome` column records 1 for that case, and 1 is right, as the next paragraph shows.

**Azerbaijan's and Bangladesh's revealed untreated outcomes (entry 22).** An untreated case reveals $Y_i(0)$, so its observed outcome and its `y0` columns must agree. In the deposit they disagree for two democratization cases, in opposite directions: Azerbaijan (1991-1992) has `outcome` 1 and `y0_obs` 0, and Bangladesh (1982-1990) has `outcome` 0 and `y0_obs` 1. The article codes the outcome 1 when authoritarianism resumes within 10 years of the transition, and Polity IV, from which the case universe was drawn, settles both: Azerbaijan's polity score falls from 1 in 1992 to -3 in 1993, and Bangladesh's holds at 6 from 1991 through 2006 and falls only in 2007. The `outcome` column is right and the `y0` columns, which the analysis reads, carry the two values transposed. Table B.2 prints the `y0` values as the deposit holds them, in both its $Y_i(0)$ and $\tilde{Y}_i(0)$ columns. **The rewrite corrects both** in a second section of `maintained/apply_appendix_c_corrections.R`, which asserts the deposited value in every column that carries the revealed $Y_i(0)$ before overwriting it, writes `output/revealed_outcome_corrections.csv`, and then asserts that the switching equation holds for every case in both samples. Both cases are left unimputed, so each enters the analysis through this value alone, and because the two values are swapped the sum of the untreated outcomes is unchanged: every table, figure, and in-text quantity the pipeline writes is

byte-identical with and without the correction. `table_b1_b2_case_dataset.R` still compares the published table against the deposit, so the comparison counts below are unaffected. Earlier versions of this report described Azerbaijan as the only case breaking the switching equation; the column that script writes had flagged Bangladesh as well.

## 5 Number-by-number comparison

Table 7: Ground truth: the published value against the deposited script and against th maintained rewrite. A blank Paper column means the quantity is not stated in th article or its appendix; a blank Archive column means no deposited script prints it.

| Location | Quantity | Paper | Archive | Rewrite |
|---|---|---|---|---|
| Table 3 | column d | 1,1,1,1,1,1,1,0,0,0 | 1,1,1,1,1,1,1,0,0,0 | 1,1,1,1,1,1,1,0,0,0 |
| Table 3 | column Y | 1,1,1,1,0,0,0,1,0,0 | 1,1,1,1,0,0,0,1,0,0 | 1,1,1,1,0,0,0,1,0,0 |
| Table 3 | column Y0_initial | ?,?,?,?,?,?,?,1,0,0 | ?,?,?,?,?,?,?,1,0,0 | ?,?,?,?,?,?,?,1,0,0 |
| Table 3 | column Y1_initial | 1,1,1,1,0,0,0,?,?,? | 1,1,1,1,0,0,0,?,?,? | 1,1,1,1,0,0,0,?,?,? |
| Table 3 | column Y0_easy | 1,1,?,?,0,?,?,1,0,0 | 1,1,?,?,0,?,?,1,0,0 | 1,1,?,?,0,?,?,1,0,0 |
| Table 3 | column Y1_easy | 1,1,1,1,0,0,0,1,0,? | 1,1,1,1,0,0,0,1,0,? | 1,1,1,1,0,0,0,1,0,? |
| Table 3 | column Y0_nulls | 1,1,?,?,0,0,0,1,0,0 | 1,1,?,?,0,0,0,1,0,0 | 1,1,?,?,0,0,0,1,0,0 |
| Table 3 | column Y1_nulls | 1,1,1,1,0,0,0,1,0,? | 1,1,1,1,0,0,0,1,0,? | 1,1,1,1,0,0,0,1,0,? |
| Table 3 | column Y0_hard | 1,1,0,0,0,0,0,1,0,0 | 1,1,0,0,0,0,0,1,0,0 | 1,1,0,0,0,0,0,1,0,0 |
| Table 3 | column Y1_hard | 1,1,1,1,0,0,0,1,0,? | 1,1,1,1,0,0,0,1,0,? | 1,1,1,1,0,0,0,1,0,? |
| Table 3 | EV bounds low, Initial Values | -40 | -40 | -40 |
| Table 3 | EV bounds high, Initial Values | 60 | 60 | 60 |
| Table 3 | EV bounds low, Easy Cases | -20 | -20 | -20 |
| Table 3 | EV bounds high, Easy Cases | 30 | 30 | 30 |
| Table 3 | EV bounds low, Nulls for 6&7 | 0 | 0 | 0 |
| Table 3 | EV bounds high, Nulls for 6&7 | 30 | 30 | 30 |
| Table 3 | EV bounds low, Hard Cases | 20 | 20 | 20 |
| Table 3 | EV bounds high, Hard Cases | 30 | 30 | 30 |
| Figure 1 | low bound, Before Any Data | -100 | -100 | -100 |
| Figure 1 | high bound, Before Any Data | 100 | 100 | 100 |
| Figure 1 | low bound, Initial Values | -40 | -40 | -40 |
| Figure 1 | high bound, Initial Values | 60 | 60 | 60 |
| Figure 1 | low bound, Easy Cases | -20 | -20 | -20 |
| Figure 1 | high bound, Easy Cases | 30 | 30 | 30 |
| Figure 1 | low bound, Nulls for 6&7 | 0 | 0 | 0 |
| Figure 1 | high bound, Nulls for 6&7 | 30 | 30 | 30 |
| Figure 1 | low bound, Hard Cases | 20 | 20 | 20 |
| Figure 1 | high bound, Hard Cases | 30 | 30 | 30 |
| Table 4 | EV bounds low, unit 6 = 0, unit 7 = 0 | 0 | 0 | 0 |
| Table 4 | EV bounds high, unit 6 = 0, unit 7 = 0 | 30 | 30 | 30 |
| Table 4 | probability, unit 6 = 0, unit 7 = 0 | 0.56 | 0.56 | 0.56 |
| Table 4 | EV bounds low, unit 6 = 1, unit 7 = 0 | -10 | -10 | -10 |
| Table 4 | EV bounds high, unit 6 = 1, unit 7 = 0 | 20 | 20 | 20 |
| Table 4 | probability, unit 6 = 1, unit 7 = 0 | 0.14 | 0.14 | 0.14 |
| Table 4 | EV bounds low, unit 6 = 0, unit 7 = 1 | -10 | -10 | -10 |
| Table 4 | EV bounds high, unit 6 = 0, unit 7 = 1 | 20 | 20 | 20 |
| Table 4 | probability, unit 6 = 0, unit 7 = 1 | 0.24 | 0.24 | 0.24 |
| Table 4 | EV bounds low, unit 6 = 1, unit 7 = 1 | -20 | -20 | -20 |
| Table 4 | EV bounds high, unit 6 = 1, unit 7 = 1 | 10 | 10 | 10 |
| Table 4 | probability, unit 6 = 1, unit 7 = 1 | 0.06 | 0.06 | 0.06 |
| Probabilistic Extension | probability-weighted point estimate, lower bound | -5 | | -5 |
| Probabilistic Extension | probability-weighted point estimate, upper bound | 25 | | 25 |
| Probabilistic Extension | 2.5th quantile of the lower bound | -20 | | -20 |
| Probabilistic Extension | 97.5th quantile of the lower bound | 0 | | 0 |
| Probabilistic Extension | 2.5th quantile of the upper bound | 10 | | 10 |
| Probabilistic Extension | 97.5th quantile of the upper bound | 30 | | 30 |
| Probabilistic Extension | probability that unit 6 has Y0 = 1 | 0.3 | 0.2 | 0.2 |
| Probabilistic Extension | probability that unit 7 has Y0 = 1 | 0.4 | 0.3 | 0.3 |

Table 7: Ground truth: the published value against the deposited script and against th maintained rewrite. A blank Paper column means the quantity is not stated in th article or its appendix; a blank Archive column means no deposited script prints i *(continued)*

| Location | Quantity | Paper | Archive | Rewrite |
|---|---|---|---|---|
| Table 5 | democratization, Negative Effect | 2% (1) | 2% (1) | 2% (1) |
| Table 5 | end of conflict, Negative Effect | | 0% (0) | 0% (0) |
| Table 5 | democratization, No Effect | 40% (25) | 40% (25) | 40% (25) |
| Table 5 | end of conflict, No Effect | | 43% (23) | 43% (23) |
| Table 5 | democratization, Positive Effect | 8% (5) | 8% (5) | 8% (5) |
| Table 5 | end of conflict, Positive Effect | | 13% (7) | 13% (7) |
| Table 5 | democratization, Unimputed | 51% (32) | 51% (32) | 51% (32) |
| Table 5 | end of conflict, Unimputed | | 44% (24) | 44% (24) |
| Figure 2 | low bound, Before Any Data | -100 | -100 | -100 |
| Figure 2 | high bound, Before Any Data | 100 | 100 | 100 |
| Figure 2 | low bound, Initial Values | -25 | -25 | -25 |
| Figure 2 | high bound, Initial Values | 75 | 75 | 75 |
| Figure 2 | low bound, Disbanded and Discredited Cases | -25 | -25 | -25 |
| Figure 2 | high bound, Disbanded and Discredited Cases | 72 | 72 | 72 |
| Figure 2 | low bound, Treated Cases | -16 | -16 | -17 |
| Figure 2 | high bound, Treated Cases | 68 | 68 | 67 |
| Figure 2 | low bound, Non-transitional Cases | -12 | -12 | -13 |
| Figure 2 | high bound, Non-transitional Cases | 62 | 62 | 61 |
| Figure 2 | low bound, Untreated Cases | -2 | -2 | -2 |
| Figure 2 | high bound, Untreated Cases | 49 | 49 | 49 |
| Figure 2 | low bound, Unimputable Cases | -2 | -2 | -2 |
| Figure 2 | high bound, Unimputable Cases | 49 | 49 | 49 |
| Figure A.1 | low bound, Before Any Data | -100 | -100 | -100 |
| Figure A.1 | high bound, Before Any Data | 100 | 100 | 100 |
| Figure A.1 | low bound, Initial Values | -33 | -33 | -33 |
| Figure A.1 | high bound, Initial Values | 67 | 67 | 67 |
| Figure A.1 | low bound, Disbanded and Discredited Cases | -32 | -32 | -32 |
| Figure A.1 | high bound, Disbanded and Discredited Cases | 61 | 61 | 61 |
| Figure A.1 | low bound, Treated Cases | -26 | -26 | -26 |
| Figure A.1 | high bound, Treated Cases | 56 | 56 | 56 |
| Figure A.1 | low bound, Non-transitional Cases | -24 | -24 | -24 |
| Figure A.1 | high bound, Non-transitional Cases | 54 | 54 | 54 |
| Figure A.1 | low bound, Untreated Cases | -3 | -3 | -3 |
| Figure A.1 | high bound, Untreated Cases | 42 | 42 | 42 |
| Figure A.1 | low bound, Unimputable Cases | -3 | -3 | -3 |
| Figure A.1 | high bound, Unimputable Cases | 42 | 42 | 42 |
| Figure 3 | low bound, ATU, Before Any Data | -100 | -100 | -100 |
| Figure 3 | high bound, ATU, Before Any Data | 100 | 100 | 100 |
| Figure 3 | low bound, ATU, Initial Values | -16 | -16 | -16 |
| Figure 3 | high bound, ATU, Initial Values | 84 | 84 | 84 |
| Figure 3 | low bound, ATU, Disbanded and Discredited Cases | -16 | -16 | -16 |
| Figure 3 | high bound, ATU, Disbanded and Discredited Cases | 80 | 80 | 80 |
| Figure 3 | low bound, ATU, Treated Cases | -16 | -16 | -16 |
| Figure 3 | high bound, ATU, Treated Cases | 80 | 80 | 80 |
| Figure 3 | low bound, ATU, Non-transitional Cases | -12 | -12 | -12 |
| Figure 3 | high bound, ATU, Non-transitional Cases | 74 | 74 | 74 |
| Figure 3 | low bound, ATU, Untreated Cases | 0 | 0 | 1 |
| Figure 3 | high bound, ATU, Untreated Cases | 58 | 58 | 60 |
| Figure 3 | low bound, ATU, Unimputable Cases | 0 | 0 | 1 |
| Figure 3 | high bound, ATU, Unimputable Cases | 58 | 58 | 60 |
| Figure 3 | low bound, ATT, Before Any Data | -100 | -100 | -100 |
| Figure 3 | high bound, ATT, Before Any Data | 100 | 100 | 100 |
| Figure 3 | low bound, ATT, Initial Values | -88 | -88 | -87.5 |
| Figure 3 | high bound, ATT, Initial Values | 12 | 12 | 12.5 |
| Figure 3 | low bound, ATT, Disbanded and Discredited Cases | -88 | -88 | -87.5 |
| Figure 3 | high bound, ATT, Disbanded and Discredited Cases | 12 | 12 | 12.5 |
| Figure 3 | low bound, ATT, Treated Cases | -15 | -15 | -22.5 |
| Figure 3 | high bound, ATT, Treated Cases | -15 | -15 | -22.5 |
| Figure 3 | low bound, ATT, Non-transitional Cases | -15 | -15 | -22.5 |

Table 7: Ground truth: the published value against the deposited script and against th maintained rewrite. A blank Paper column means the quantity is not stated in th article or its appendix; a blank Archive column means no deposited script prints i *(continued)*

| Location | Quantity | Paper | Archive | Rewrite |
|---|---|---|---|---|
| Figure 3 | high bound, ATT, Non-transitional Cases | -15 | -15 | -22.5 |
| Figure 3 | low bound, ATT, Untreated Cases | -15 | -15 | -22.5 |
| Figure 3 | high bound, ATT, Untreated Cases | -15 | -15 | -22.5 |
| Figure 3 | low bound, ATT, Unimputable Cases | -15 | -15 | -22.5 |
| Figure 3 | high bound, ATT, Unimputable Cases | -15 | -15 | -22.5 |
| Figure A.2 | low bound, ATU, Before Any Data | -100 | -100 | -100 |
| Figure A.2 | high bound, ATU, Before Any Data | 100 | 100 | 100 |
| Figure A.2 | low bound, ATU, Initial Values | -29 | -29 | -29 |
| Figure A.2 | high bound, ATU, Initial Values | 71 | 71 | 71 |
| Figure A.2 | low bound, ATU, Disbanded and Discredited Cases | -27 | -27 | -27.5 |
| Figure A.2 | high bound, ATU, Disbanded and Discredited Cases | 64 | 64 | 64 |
| Figure A.2 | low bound, ATU, Treated Cases | -27 | -27 | -27.5 |
| Figure A.2 | high bound, ATU, Treated Cases | 64 | 64 | 64 |
| Figure A.2 | low bound, ATU, Non-transitional Cases | -26 | -26 | -26 |
| Figure A.2 | high bound, ATU, Non-transitional Cases | 62 | 62 | 62 |
| Figure A.2 | low bound, ATU, Untreated Cases | -2 | -2 | -2 |
| Figure A.2 | high bound, ATU, Untreated Cases | 48 | 48 | 48 |
| Figure A.2 | low bound, ATU, Unimputable Cases | -2 | -2 | -2 |
| Figure A.2 | high bound, ATU, Unimputable Cases | 48 | 48 | 48 |
| Figure A.2 | low bound, ATT, Before Any Data | -100 | -100 | -100 |
| Figure A.2 | high bound, ATT, Before Any Data | 100 | 100 | 100 |
| Figure A.2 | low bound, ATT, Initial Values | -67 | -67 | -67 |
| Figure A.2 | high bound, ATT, Initial Values | 33 | 33 | 33 |
| Figure A.2 | low bound, ATT, Disbanded and Discredited Cases | -67 | -67 | -67 |
| Figure A.2 | high bound, ATT, Disbanded and Discredited Cases | 33 | 33 | 33 |
| Figure A.2 | low bound, ATT, Treated Cases | -10 | -10 | -10 |
| Figure A.2 | high bound, ATT, Treated Cases | -10 | -10 | -10 |
| Figure A.2 | low bound, ATT, Non-transitional Cases | -10 | -10 | -10 |
| Figure A.2 | high bound, ATT, Non-transitional Cases | -10 | -10 | -10 |
| Figure A.2 | low bound, ATT, Untreated Cases | -10 | -10 | -10 |
| Figure A.2 | high bound, ATT, Untreated Cases | -10 | -10 | -10 |
| Figure A.2 | low bound, ATT, Unimputable Cases | -10 | -10 | -10 |
| Figure A.2 | high bound, ATT, Unimputable Cases | -10 | -10 | -10 |
| Figure 4 | low bound, us, The 20 Cases with Expert Responses | -10 | -10 | -10 |
| Figure 4 | high bound, us, The 20 Cases with Expert Responses | 40 | 40 | 40 |
| Figure 4 | low bound, expert, The 20 Cases with Expert Responses | -15 | -15 | -15 |
| Figure 4 | high bound, expert, The 20 Cases with Expert Responses | 15 | 15 | 15 |
| Figure 4 | low bound, us, All 63 Democratization Cases | -2 | -2 | -2 |
| Figure 4 | high bound, us, All 63 Democratization Cases | 49 | 49 | 49 |
| Figure 4 | low bound, expert, All 63 Democratization Cases | -19 | -19 | -19 |
| Figure 4 | high bound, expert, All 63 Democratization Cases | 59 | 59 | 59 |
| Figure 4 | low bound, combined, All 63 Democratization Cases | -3 | -3 | -3 |
| Figure 4 | high bound, combined, All 63 Democratization Cases | 41 | 41 | 41 |
| Abstract | bounds width prior to any analysis | 100 | 100 | 100 |
| Abstract | bounds width after imputation | 51 | 51 | 51 |
| Abstract | bounds width after the expert survey | 44 | 44 | 44 |
| Application to the Average Effect of Truth Commissions | democratization cases | 63 | 63 | 63 |
| Application to the Average Effect of Truth Commissions | treated democratization cases | 8 | 8 | 8 |
| Application to the Average Effect of Truth Commissions | bounds width before any data are collected | 200 | 200 | 200 |
| Application to the Average Effect of Truth Commissions | bounds width once half the potential outcomes are revealed | 100 | 100 | 100 |
| Application to the Average Effect of Truth Commissions | final ATE lower bound | -2 | -2 | -2 |

Table 7: Ground truth: the published value against the deposited script and against th maintained rewrite. A blank Paper column means the quantity is not stated in th article or its appendix; a blank Archive column means no deposited script prints i *(continued)*

| Location | Quantity | Paper | Archive | Rewrite |
|---|---|---|---|---|
| Application to the Average Effect of Truth Commissions | final ATE upper bound | 49 | 49 | 49 |
| Application to the Average Effect of Truth Commissions | final ATE bounds width | 51 | 51 | 51 |
| A Toy Example | toy example: treated units with outcome 1 | 3 | 4 | 4 |
| A Toy Example | toy example: untreated units with outcome 1 | 1 | 1 | 1 |
| A Toy Example | toy example: bounds width before any data | 200 | 200 | 200 |
| A Toy Example | toy example: final bounds width | 10 | 10 | 10 |
| Application to the Average Effect of Truth Commissions | ATU final bounds width | 58 | 58 | 58 |
| Application to the Average Effect of Truth Commissions | ATT summarised as a point effect | -15 | -15 | -22.5 |
| Application to the Average Effect of Truth Commissions | cases imputed as a non-zero causal effect | 6 | 6 | 6 |
| Expert Survey | expert responses received | 20 | 20 | 20 |
| Expert Survey | democratization cases without an expert response | 43 | 43 | 43 |
| Application to the Average Effect of Truth Commissions | our bounds width, 20 responding cases | 50 | 50 | 50 |
| Expert Survey | expert bounds width, 20 responding cases | 30 | 30 | 30 |
| Application to the Average Effect of Truth Commissions | our bounds width, all 63 cases | 51 | 51 | 51 |
| Expert Survey | expert bounds width, all 63 cases | 78 | 78 | 78 |
| Application to the Average Effect of Truth Commissions | combined bounds width, all 63 cases | 44 | 44 | 44 |
| Expert Survey | expert and author imputations agreed | 7 | 7 | 7 |
| Expert Survey | expert and author imputations conflicted | 3 | 3 | 3 |
| Expert Survey | we imputed, the experts did not | 3 | 3 | 3 |
| Expert Survey | the experts imputed, we did not | 7 | 7 | 7 |
| Expert Survey | cases the experts imputed | 14 | 14 | 14 |
| Application to the Average Effect of Truth Commissions | cases we imputed | 10 | 10 | 10 |
| Expert Survey | experts who declined to impute | 6 | 6 | 6 |
| Expert Survey | declining experts assigned to an untreated case | 6 | 6 | 6 |
| Expert Survey | expert imputations differing from the observed outcome | 0 | 0 | 0 |
| Appendix A | end-of-conflict cases | 54 | 54 | 54 |
| Appendix A | treated end-of-conflict cases | 6 | 6 | 6 |
| Appendix A | bounds width once half the potential outcomes are revealed | 100 | 100 | 100 |
| Appendix A | bounds width after imputation | 41 | 44 | 44 |
| Appendix A | final ATE lower bound | -3 | -3 | -3 |
| Appendix A | final ATE upper bound | 42 | 42 | 42 |
| Appendix A | final ATE bounds width from the rounded endpoints | 45 | 45 | 45 |
| Appendix A | ATT summarised as a point effect | -10 | -10 | -10 |
| A Toy Example | toy example: units | 10 | | 10 |
| A Toy Example | toy example: treated units | 7 | | 7 |
| A Toy Example | toy example: untreated units | 3 | | 3 |
| A Toy Example | toy example: easy imputations | 5 | | 5 |
| A Toy Example | toy example: untreated outcomes for units 1, 2 and 5 | 1, 1, 0 | | 1, 1, 0 |
| A Toy Example | toy example: treated outcomes for units 8 and 9 | 1, 0 | | 1, 0 |
| A Toy Example | probabilistic extension: possible sets of potential outcomes | 4 | | 4 |
| Imputing Missing Potential Outcomes | unobserved potential outcomes | 63 | | 63 |
| Imputing Missing Potential Outcomes | unobserved potential outcomes imputed with certainty | 5 | | 3 |
| Imputing Missing Potential Outcomes | unobserved potential outcomes imputed probabilistically | 26 | | 28 |
| Imputing Missing Potential Outcomes | unobserved potential outcomes left unimputed | 32 | | 32 |
| Imputing Missing Potential Outcomes | disbanded and discredited democratization cases | 4 | | 2 |
| Imputing Missing Potential Outcomes | observed outcome in the disbanded and discredited cases | 0 | | 0 |
| Imputing Missing Potential Outcomes | South Africa: probability the imputed untreated outcome equals 1 | 0.8 | | 0.8 |
| Imputing Missing Potential Outcomes | disbanded cases: probability the imputed outcome equals 1 | 0.1 | | 0.1 |
| Imputing Missing Potential Outcomes | Nigeria: probability the imputed untreated outcome equals 1 | 0.1 | | 0.1 |
| Expert Survey | experts disagreeing with our coding of treatment and outcome | 5 | | |
| Discussion | expert imputations conflicting with ours, restated in the discussion | 3 | | 3 |

Table 7: Ground truth: the published value against the deposited script and against th maintained rewrite. A blank Paper column means the quantity is not stated in th article or its appendix; a blank Archive column means no deposited script prints i *(continued)*

| Location | Quantity | Paper | Archive | Rewrite |
|---|---|---|---|---|
| Application to the Average Effect of Truth Commissions | end-of-conflict cases, stated in the main text footnote | 54 | | 54 |
| Application to the Average Effect of Truth Commissions | unimputed cases as a share of the total | | | 32 of 63, 50.8% |
| Application to the Average Effect of Truth Commissions | the bounds around the ATE include zero | | | [-2, 49] contains 0 |
| Application to the Average Effect of Truth Commissions | the ATT bounds shrink to a point | | | [-22.5, -22.5], width |
| Application to the Average Effect of Truth Commissions | the ATU bounds are wider than the ATT bounds | | | 58 against 0 |
| Application to the Average Effect of Truth Commissions | the combined bounds are narrower than our original bounds | | | 44 against 51 |
| Expert Survey | no expert assigned to a treated case declined to impute | | | 0 of 6 declining exp |
| Expert Survey | expert bounds are narrower than ours among the 20 responding cases | | | 30 against 50 |
| Expert Survey | expert bounds are wider than ours among all 63 cases | | | 78 against 51 |
| Table B.1 | cells printed in the treatment indicator column | 54 | | 54 |
| Table B.1 | cells printed in the observed outcome column | 54 | | 54 |
| Table B.1 | cells printed in the revealed potential outcome column | 54 | | 54 |
| Table B.1 | cells printed in the imputed untreated outcome column | 54 | | 54 |
| Table B.1 | cells printed in the imputed treated outcome column | 54 | | 54 |
| Table B.1 | cells printed in the imputation probability column | 30 | | 30 |
| Table B.1 | cells printed in the unit-level effect column | 54 | | 54 |
| Table B.2 | cells printed in the treatment indicator column | 63 | | 63 |
| Table B.2 | cells printed in the observed outcome column | 63 | | 62 |
| Table B.2 | cells printed in the revealed potential outcome column | 63 | | 63 |
| Table B.2 | cells printed in the imputed untreated outcome column | 63 | | 63 |
| Table B.2 | cells printed in the imputed treated outcome column | 63 | | 63 |
| Table B.2 | cells printed in the imputation probability column | 31 | | 30 |
| Table B.2 | cells printed in the unit-level effect column | 63 | | 63 |

Of the 232 recorded claims, 181 can be compared against both the article and a deposited script, and 177 of those match. Six further claims are stated in the article but computed by no deposited script: the probability-weighted point estimate of the bounds in the probabilistic extension, [-5, 25], and the 2.5th and 97.5th quantiles of each bound, [-20, 0] and [10, 30]. The rewrite computes all six from the enumerated scenarios in `table_4_probabilistic.R` and all six match.

## 5.1 Coverage

The list of numbers to check is built from the article rather than from the pipeline, because a list built from the pipeline is silent exactly where the pipeline is missing something. `ground_truth/published_claims.csv` is every numeric token the article and its supporting information print, 268 of them, each classified by hand. 228 are quantities this pipeline can move, and each of those must be checked twice: once by a row in the ground truth and once by

a block in `maintained/in_text_claims.R`, which reaches the same number by its own path through the same output files. The remaining 40 are scale endpoints, unit indices, period ranges and values copied out of other authors' papers, none of which any analysis can change.

Table 8: The published claims extraction, by class.

| Class | Claims | Checked by |
|---|---|---|
| definitional | 22 | at the point of use |
| descriptive | 9 | ground truth and in_text_claims.R |
| pipeline | 219 | ground truth and in_text_claims.R |
| structural | 14 | at the point of use |
| transcribed | 4 | at the point of use |

`ground_truth/build_ground_truth.R` enforces this. It sources the claims file rather than reading it as text, because a block that errors or prints nothing satisfies a textual check completely, and it counts the printed claims against the extraction. It also compares the two instruments value by value: they filter, convert and round independently, so a disagreement between them means one of the two is wrong. The run halts on any of these.

### 5.2 Appendix Tables B.1 and B.2

Appendix B prints the full case datasets, 54 end-of-conflict cases and 63 democratization cases, with seven columns each. The deposited archive has no script for either table, so until `table_b1_b2_case_dataset.R` was written the 763 published cells had nothing on this side to be compared against. All but 2 reproduce; both exceptions are described in the errata section above. That script is the one place in the rewrite that reads the deposit rather than the corrected case file, because the question it answers is whether the published pages print what the deposit holds. It also writes `table_b2_corrected_rows.csv`, the 7 rows of Table B.2 as published beside the same rows with the appendix C probabilities restored.

## 6 Maintained rewrite

The rewrite lives in `maintained/`: a helpers file, a corrections script, one cleaning script, four table scripts, four figure scripts and five in-text scripts. It is a translation rather than a reanalysis in every respect but two. `ev_bounds()`, `bounds_width()`, `sample_bounds()` and the probabilistic-extension functions are the paper's contribution and are carried over with their logic unchanged; what changes is the tidyverse around them. The first exception is `apply_appendix_c_corrections.R`, which restores the 7 imputation probabilities the deposit transcribes differently from the appendix that states them, and corrects the two revealed untreated outcomes the deposit transposes between Azerbaijan and Bangladesh. The second is that the four bounds figures enumerate the distribution of each bound with `bounds_distribution()`

rather than sampling it with `sample_bounds()`, which is the same quantity computed exactly instead of estimated. Both are described in the sections above.

## 6.1 Architecture

Table 9: The maintained rewrite.

| Script | Output |
|---|---|
| helpers.R | none; packages, the QUIMPO functions and the toy example |
| apply_appendix_c_corrections.R | cases_corrected.rds, appendix_c_corrections.csv |
| clean_cases.R | cases_long.rds |
| table_3_toy_example.R | table_3_toy_example.csv, table_3_bounds.csv |
| table_4_probabilistic.R | table_4_probabilistic.csv, table_4_summary.csv |
| table_5_imputation_summary.R | table_5_imputation_summary.csv |
| table_b1_b2_case_dataset.R | table_b1_b2_case_dataset.csv, table_b2_corrected_rows.csv |
| figure_1_toy_bounds.R | figure_1_toy_bounds.pdf/.png/.csv |
| figures_2_a1_ate_bounds.R | figure_2_dem_ate_bounds.pdf/.png, figure_a1_eoc_ate_bounds.pdf/.png, figures_2_a1_gg_df.csv |
| figures_3_a2_att_atu_bounds.R | figure_3_dem_att_atu_bounds.pdf/.png, figure_a2_eoc_att_atu_bounds.pdf/.png, figures_3_a2_gg_df.csv |
| figure_4_expert_validation.R | figure_4_expert_validation.pdf/.png/.csv |
| text_expert_agreement.R | text_expert_agreement.csv, text_expert_agreement_counts.csv |
| text_imputation_counts.R | text_imputation_counts.csv |
| text_summary_stats.R | text_summary_stats.csv |
| text_expected_bounds.R | text_expected_bounds.csv |
| text_seed_sensitivity.R | text_seed_sensitivity.csv |
| in_text_claims.R | none; one printed line per published claim, checked by build_ground_truth.R |

Three departures from a line-by-line translation are worth naming.

The toy example is defined once, in `quimpo_toy_example()` in `helpers.R`, rather than retyped in each of the three scripts that need it. In the archive, `table_3_figure_1.R` and `table_4.R` each set up the ten units and walk them through the imputation steps independently, which is the arrangement in which Table 3, Figure 1 and Table 4 can silently come to describe different examples.

Figures 2 and A1 come from one script and Figures 3 and A2 from another, because in each pair the two figures are two facets of one computation and splitting them would mean doing it twice. The archive's `figures_2_3_A1_A2.R` puts all four in one file; the rewrite splits along that boundary rather than the file boundary.

`text_summary_stats.R` reads the bounds back out of the figure output rather than recomputing them, so the widths quoted in the abstract and the body cannot drift from the figures they summarise. Every number it prints is traceable to a committed CSV, and no published value is typed into any script in `maintained/`.

## 6.2 Deprecated patterns replaced

Table 10: Deprecated patterns and their replacements in the maintained rewrite.

| Original pattern | Replacement |
|---|---|
| ‘rm(list = ls())‘ | (omitted) |
| commented-out ‘setwd()‘ | ‘here::here()‘ |
| magrittr pipe | native pipe |
| ‘reshape2::melt()‘ / ‘dcast()‘ | ‘pivot_longer()‘ / ‘pivot_wider()‘ |
| ‘do(sample_bounds(...))‘ | ‘reframe(sample_bounds(...))‘ |
| ‘map_df(enframe, .id = )‘ | ‘map(enframe) |> bind_rows(.id = )‘ |
| ‘geom_errorbarh(height = )‘ | ‘geom_errorbar(orientation = ”y”, width = )‘ |
| ‘geom_linerange(size = )‘ | ‘geom_linerange(linewidth = )‘ |
| ‘<ggplot> %+% data‘ | a named plot function taking the data as an argument |
| ‘separate()‘ | ‘separate_wider_delim()‘ |
| ‘recode_factor()‘ | ‘case_when()‘ + ‘factor()‘ |
| ‘xtable‘ + ‘print.xtable‘ (commented out) | ‘write_csv()‘ to ‘output/‘ |
| ‘ggsave()‘ (commented out) | ‘ggsave()‘ to ‘output/‘, PDF and PNG |
| unseeded ‘rbinom()‘ simulation of the bounds | exact enumeration of the same distribution |
| bounds printed as fractions in ‘table_4.R‘ | multiplied by 100, as the paper prints them |

# 7 Figures

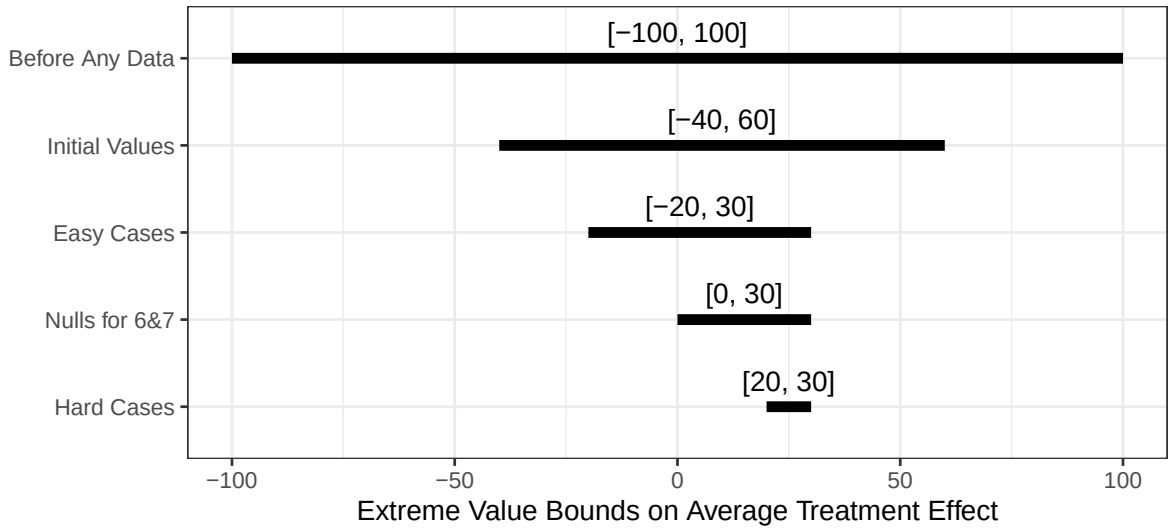



Figure 1: Figure 1: extreme value bounds at each step of the toy example.

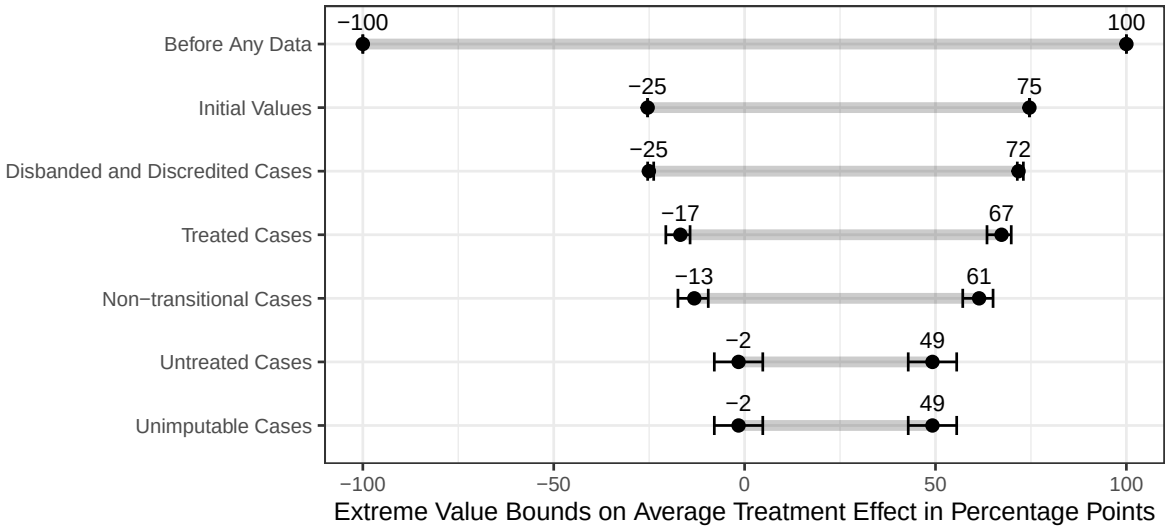



Figure 2: Figure 2: transitional truth commissions, extreme value bounds on the ATE.

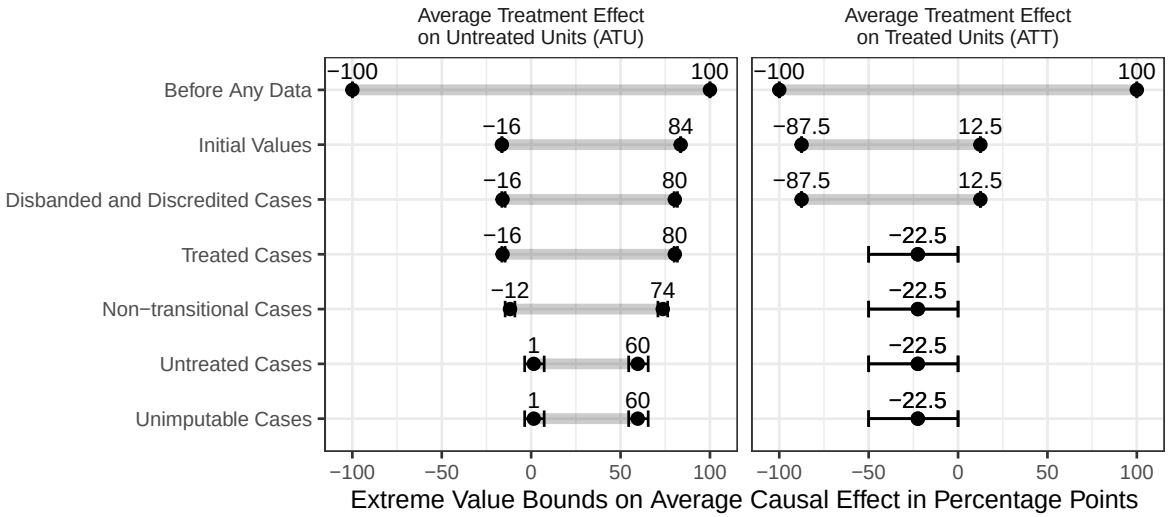



Figure 3: Figure 3: extreme value bounds by treatment status.

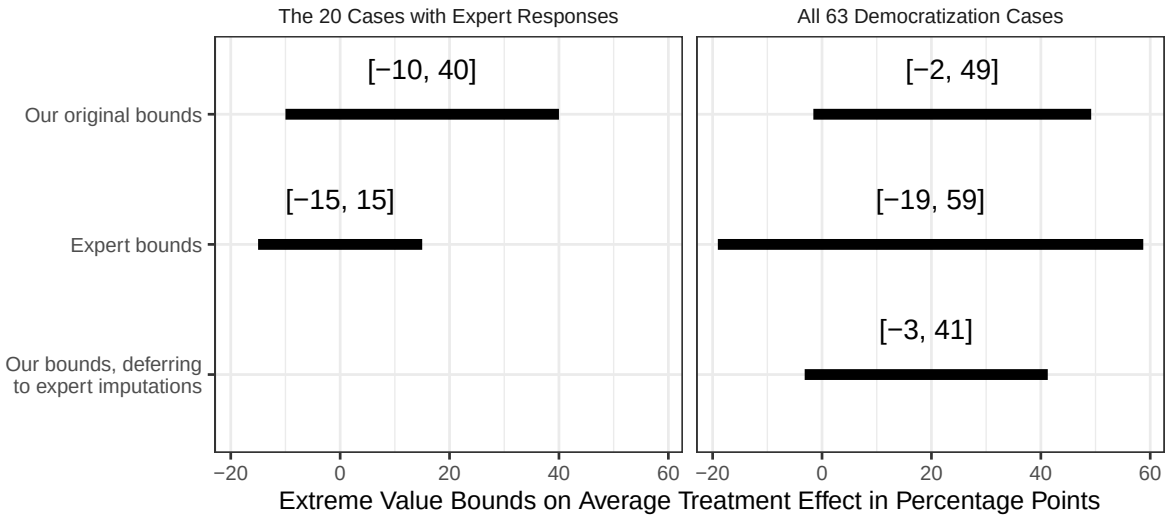



Figure 4: Figure 4: extreme value bounds implied by the expert imputations.

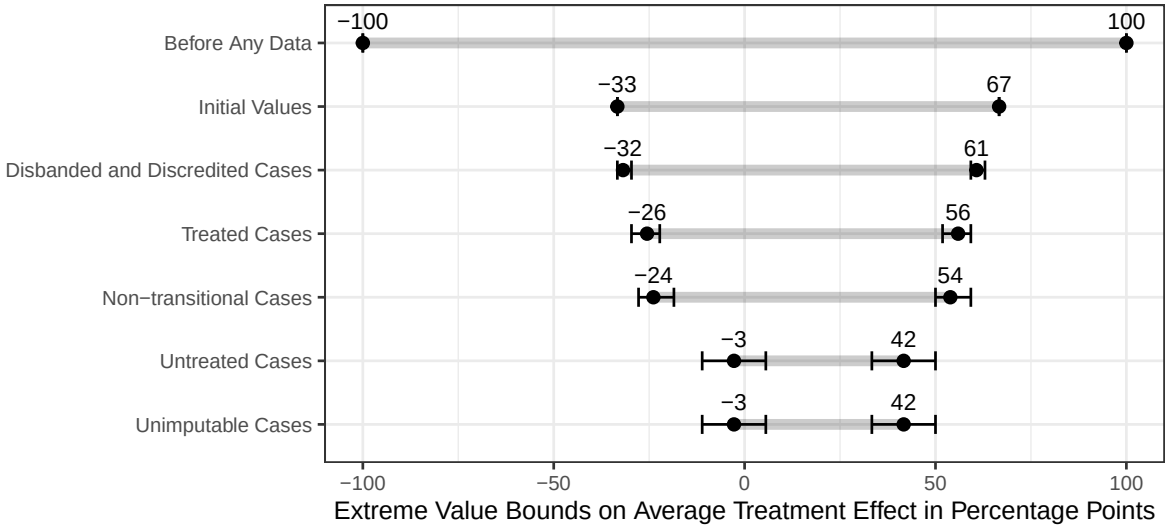



Figure 5: Figure A1: end-of-conflict cases, extreme value bounds on the ATE.

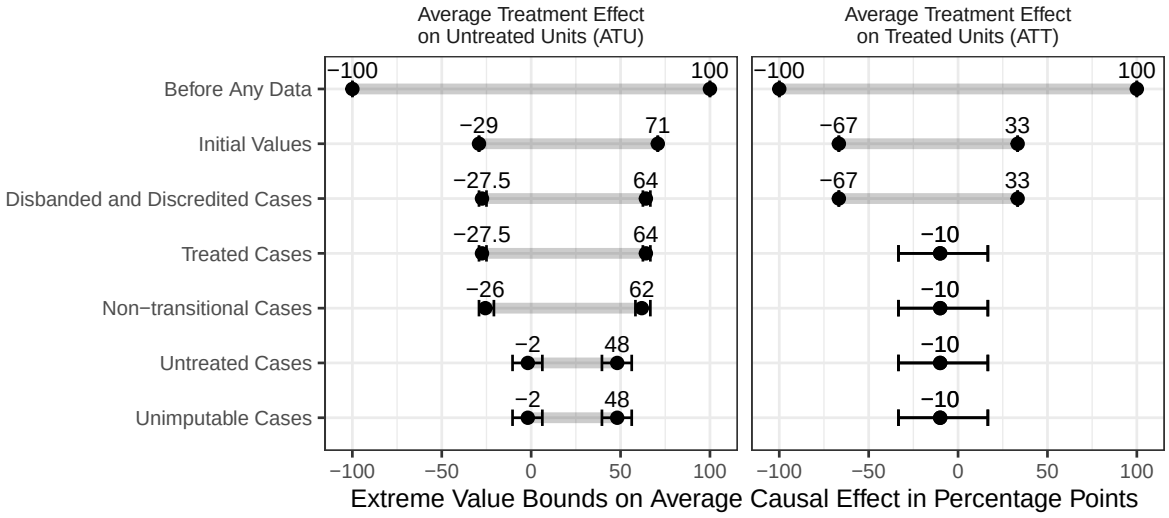



Figure 6: Figure A2: end-of-conflict cases, extreme value bounds by treatment status.

## 8 Expert survey verification

The Expert Survey section carries eleven quantities that no deposited script writes to disk, three of them recorded only in a code comment in `figure_4.R`. `text_expert_agreement.R` recomputes all of them from the deposited case data.

Table 11: The Expert Survey section, recomputed. Agreement means the two sides made the same imputation or both declined; the last row checks the claim that all 20 respondents either declined or imputed the observed outcome.

| Quantity | Rewrite | Paper |
|---|---|---|
| agreement | 7 | 7 |
| disagreement | 3 | 3 |
| They declined, we imputed | 3 | 3 |
| They imputed, we declined | 7 | 7 |
| expert responses received | 20 | 20 |
| democratization cases without an expert response | 43 | 43 |
| cases the experts imputed | 14 | 14 |
| cases we imputed | 10 | 10 |
| experts who declined to impute | 6 | 6 |
| declining experts assigned to an untreated case | 6 | 6 |
| expert imputations differing from the observed outcome | 0 | 0 |

The archive's `figure_4.R` computes the agreement coding and prints it with `table()`, then records the answer in three comment lines: "uruguay, burundi, south africa are 'disagreements' ",

"7 cases, they imputed when we did not", "3 cases, we imputed and they did not". All three are correct. They are also the only place the numbers exist, which is the pattern this program treats as a defect regardless of whether the comment is right: a comment is not an output, and nothing checks it.

## 9 In-text quantities

Table 12: Bounds widths and summary effects quoted in the abstract, the body and Appendix A, read back out of the figure output that produced them.

| Quantity | Rewrite | As a figure labels it |
|---|---|---|
| dem: bounds width before any data | 200.00 | 200 |
| dem: bounds width once the world reveals half the potential outcomes | 100.00 | 100 |
| dem: final ATE lower bound | -1.59 | -2 |
| dem: final ATE upper bound | 49.21 | 49 |
| dem: final ATE bounds width | 50.79 | 51 |
| dem: ATU final bounds width | 58.18 | 58 |
| dem: ATT final lower bound | -22.50 | -22.5 |
| dem: ATT final upper bound | -22.50 | -22.5 |
| expert: our bounds width, 20 responding cases | 50.00 | 50 |
| expert: expert bounds width, 20 responding cases | 30.00 | 30 |
| expert: our bounds width, all 63 cases | 50.79 | 51 |
| expert: expert bounds width, all 63 cases | 77.78 | 78 |
| expert: combined bounds width, all 63 cases | 44.44 | 44 |
| dem: cases imputed as a non-zero causal effect | 6.00 | 6 |
| eoc: bounds width once the world reveals half the potential outcomes | 100.00 | 100 |
| eoc: final ATE lower bound | -2.78 | -3 |
| eoc: final ATE upper bound | 41.67 | 42 |
| eoc: final ATE bounds width | 44.44 | 44 |
| eoc: ATT final lower bound | -10.00 | -10 |
| eoc: ATT final upper bound | -10.00 | -10 |

The abstract and the body describe the same starting point in two ways, and both are right. The body says the bounds are 200 points wide "before any data are collected" and shrink to 100 "after the world reveals half the potential outcomes". The abstract skips the first of those and says the bounds are 100 points wide "prior to any analysis", meaning after data collection and before imputation. Both numbers are in the pipeline and both reproduce.

## 10 R environment

Table 13: Package versions used for this report.

| Package | Version |
|---|---|

| | |
|---|---|
| tidyverse | 2.0.0 |
| here | 1.0.2 |
| janitor | 2.2.1 |
| knitr | 1.52 |
| kableExtra | 1.4.1 |
| ggplot2 | 4.0.3 |
| dplyr | 1.2.1 |
| tidyr | 1.3.2 |

R version: R version 4.6.2 (2026-10-29). The archive was written against R 4.1.2.